

Writing and Solving Linear Inequalities

AQA · KS4 Year 9–10

Name: _____ Class: _____ Date: _____

Write inequalities from descriptions and solve linear inequalities. Use the same balancing steps as for equations, but remember to reverse the inequality sign whenever you multiply or divide both sides by a negative number.

VISUAL EXAMPLE



$x < 5$ on a number line: open circle at 5, shade everything to the left.

COMMON MISTAKES

Forgetting to reverse the sign when multiplying or dividing by a negative number.

Reversing the sign when only adding or subtracting — the sign only flips for $\times$ or $\div$ by a negative.

Treating $<$ and $\leq$ as the same: $\leq$ includes the end value, $<$ does not.

Writing the answer as an equation ($x = \dots$) instead of an inequality ($x < \dots$).

METHOD

- 1 Treat the inequality like an equation — keep both sides balanced.
- 2 Use inverse operations to undo $+/-$ first, then $\times/\div$.
- 3 If you multiply or divide both sides by a negative, REVERSE the inequality sign.
- 4 Write the solution as an inequality, e.g. $x > 4$.
- 5 Check by testing a value from your solution set in the original inequality.

WORKED EXAMPLE

Solve: $3x + 4 < 19$

Step 1 — Subtract 4 from both sides: $3x < 15$.

Step 2 — Divide both sides by 3 (positive, so sign stays): $x < 5$.

Step 3 — No negative division, so no flip needed.

Step 4 — Solution: $x < 5$.

Step 5 — Check $x = 4$: $3(4) + 4 = 16$, and $16 < 19$

1 Solve.

(2 marks)

a) $x + 3 > 7$

b) $x - 2 \leq 5$

HOW TO START

Solve like an equation: do the same to both sides.

For $x + 3 > 7$, subtract 3 from both sides: $x > 4$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Balance both sides like an equation | 2 Undo +/- first, then $\times/\div$ | 3 Flip the sign if $\times$ or $\div$ by a negative | 4 Write the answer as an inequality | 5 Check with a value from the solution

2 Solve.

(2 marks)

a) $2x < 10$

b) $5x \geq 20$

HOW TO START

Divide both sides by the number in front of x .

For $2x < 10$, divide by 2: $x < 5$. The sign stays the same (dividing by a positive).

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Balance both sides like an equation | 2 Undo $\pm$ first, then $\times/\div$ | 3 Flip the sign if $\times$ or $\div$ by a negative | 4 Write the answer as an inequality | 5 Check with a value from the solution

3 Solve.

(3 marks)

a) $2x + 1 > 9$

b) $3x - 4 \leq 11$

c) $4x + 5 < 5$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Balance both sides like an equation | 2 Undo +/- first, then $\times/\div$ | 3 Flip the sign if $\times$ or $\div$ by a negative | 4 Write the answer as an inequality | 5 Check with a value from the solution

4 Solve. Remember to reverse the sign when dividing by a negative.

(3 marks)

a) $-x > 3$

b) $-2x \geq 8$

c) $5 - x < 2$

ANSWERS

(a) $=$ _____

(b) $=$ _____

(c) $=$ _____

METHOD 1 Balance both sides like an equation | 2 Undo +/- first, then $\times/\div$ | 3 Flip the sign if $\times$ or $\div$ by a negative | 4 Write the answer as an inequality | 5 Check with a value from the solution

5 Solve.

(3 marks)

a) $3(x - 2) = 9$

b) $2(x + 1) > x + 5$

c) $4x - 7 = 2x + 3$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Balance both sides like an equation | 2 Undo +/- first, then $\times/\div$ | 3 Flip the sign if $\times$ or $\div$ by a negative | 4 Write the answer as an inequality | 5 Check with a value from the solution

6

(3 marks)

a) Sam has £ x . After spending £12, Sam has less than £20 left. (Write and solve.)

b) A taxi ride costs £4 plus £2 per mile and must cost no more than £20. What is the greatest number of whole miles?

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Balance both sides like an equation | 2 Undo $\pm$ first, then $\times \div$ | 3 Flip the sign if $\times$ or $\div$ by a negative | 4 Write the answer as an inequality | 5 Check with a value from the solution

(3 marks)

b) Solve $2x - 5 < 7$ and write the largest integer value of x . (1 mark)

(a) =

(b) =

METHOD 1 Balance both sides like an equation | 2 Undo $\pm$ first, then $\times/\div$ | 3 Flip the sign if $\times$ or $\div$ by a negative | 4 Write the answer as an inequality | 5 Check with a value from the solution

8 A rectangle has width x cm and length $(x + 4)$ cm, where x is a positive integer. (5 marks)
The perimeter is less than 36 cm.

- a) Form an inequality in x . (1 mark)
- b) Solve it. (2 marks)
- c) List all the possible integer values of x . (2 marks)

ANSWERS

(a) = _____
(b) = _____
(c) = _____

METHOD 1 Balance both sides like an equation | 2 Undo $\pm$ first, then $\times/\div$ | 3 Flip the sign if $\times$ or $\div$ by a negative | 4 Write the answer as an inequality | 5 Check with a value from the solution

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet

Getting there

Confident

Really confident

Which question did you find the hardest, and why?

When exactly do I reverse the inequality sign?

One thing I will practise before the next lesson:

Teacher key

Q1: a) $x > 4$ b) $x \leq 7$

Q2: a) $x < 5$ b) $x \geq 4$

Q3: a) $2x > 8 \Rightarrow x > 4$ b) $3x \leq 15 \Rightarrow x \leq 5$ c) $4x < 0 \Rightarrow x < 0$

Q4: a) $-x > 3 \Rightarrow x < -3$ b) $-2x \geq 8 \Rightarrow x \leq -4$ (sign flips) c) $5 - x < 2 \Rightarrow -x < -3 \Rightarrow x > 3$

Q5: a) $3x - 6 \leq 9 \Rightarrow 3x \leq 15 \Rightarrow x \leq 5$ b) $2x + 2 > x + 5 \Rightarrow x > 3$ c) $2x \leq 10 \Rightarrow x \leq 5$

Q6: a) $x - 12 < 20 \Rightarrow x < 32$ b) $4 + 2m \leq 20 \Rightarrow 2m \leq 16 \Rightarrow m \leq 8$, so 8 whole miles

Q7: a) $4x \leq 16 \Rightarrow x \leq 4$, smallest integer = 4 b) $2x < 12 \Rightarrow x < 6$, largest integer = 5

Q8: a) $2(x + x + 4) < 36$, i.e. $4x + 8 < 36 \Rightarrow 4x < 28 \Rightarrow x < 7$ c) $x = 1, 2, 3, 4, 5, 6$
